



FLEET & SHORE PERSONNEL TRAINING

Human Factors Management and Self Assessment

Understanding human factors, the five pillars of human performance, and the self-assessment KPIs that measure them.

OCIMF Element 14

UK MCA Deadly Dozen

ISM Code / SMS

Company SMS

What we will cover

01

Introduction to Human Factors

Module 1 — concept, disasters, human error and human failure

02

The Human Factors Approach

Module 2 — definitions, principles and the five pillars

03

Self-Assessment: Policy and Capability

Module 3 — maturity levels and KPIs 14.1 to 14.4

04

Self-Assessment: Design, Skills, Learning

Module 3 — the remaining three KPI groups

05

Human Factors in the Management System

Module 4 — SMS elements and error management culture



SECTION 01

Introduction to Human Factors

Human factors is a key enabler to further reduce safety, environment, security and health impacts within our industry.

THE OCIMF POSITION

Human factors has to be part of any management system

Human factors is an integral part of the OCIMF strategy and a key enabler to further reduce safety, environment, security and health impacts within our industry. It cannot be implemented overnight — it takes time for companies to become familiar with the concepts and understand how to apply them practically.

Integral to strategy

Part of the SMS

Built over time

What the industry is asking of companies

The intention is to integrate a human factors element into the suite of management self assessments, and feedback from users is very much encouraged.



Make it part of the management system

Human factors is not a separate initiative. It belongs inside the SMS, alongside procedures, risk assessment and training.



Accept that it takes time

A human factors element cannot be implemented overnight. Companies need time to become familiar with the concepts.



Learn to apply it practically

Understanding the theory is not enough — the concepts have to be translated into tasks, equipment and daily decisions.



Measure it with self assessment

The human factors element is being integrated into the suite of management self assessments so progress can be measured.



Feed back what works

Feedback from users is actively encouraged, so that guidance improves with real shipboard and shore experience.



A key enabler

Human factors is described as a key enabler to further reduce safety, environment, security and health impacts. It is a means of improving performance, not an additional compliance burden.

Human factors is the individual and the context

Human factors looks at the person and at everything around the person. Performance is the product of both.



Individual

What the person brings

- Physical — strength, fitness, senses and physical limits
- Psychological — attention, memory, stress and motivation
- Social — relationships, culture and communication
- Competency — knowledge, skill and experience



Context

What surrounds the person

- Equipment — design, condition and availability
- Procedures — clarity, accuracy and usability
- Team — roles, supervision and support
- Other individuals — everyone whose actions interact with theirs

Major disasters with human factors at their core

These events are studied across industries because in each case the conditions around people, not simply individual failure, shaped the outcome.



Germanwings

Aviation. Highlighted mental health, monitoring and organisational responsibility.



Exxon Valdez

Grounding and major oil spill. Fatigue, manning and watchkeeping conditions.



Costa Concordia

Passenger ship grounding. Leadership, deviation from plan and bridge culture.



Deepwater Horizon

Blowout and loss of life. Safety-critical decisions under commercial pressure.

What is easy to see, and what is not

An incident is only the tip. Everything that produced it sits below the waterline.



Easy to see

Above the waterline

- The incident itself
- Observable behaviour
- The event as reported



Below the waterline

The real subject matter

- Human performance
- Work as done, not work as imagined
- Safety culture



Difficult to see

What has to be looked for

- Workload, fatigue, situation awareness, stress and the interaction among SHEL
- System demands, workarounds, internal and external targets
- Norms, values, perceptions and organisational culture

HUMAN ERROR

After decades of development, human error is still the main cause of incidents

Human error remains the most common single factor identified in incidents and accidents. The useful question is not who erred, but why the conditions made that error likely — and what can be changed.

Ask why, not who

Change the conditions

Learn, do not blame

The Deadly Dozen — people make mistakes

The Deadly Dozen describes twelve common people-related factors that affect the likelihood of critical mistakes.



Traps, not faults

They are traps that can lead to an unintended direction or to harm — conditions that catch competent people.



Twelve common factors

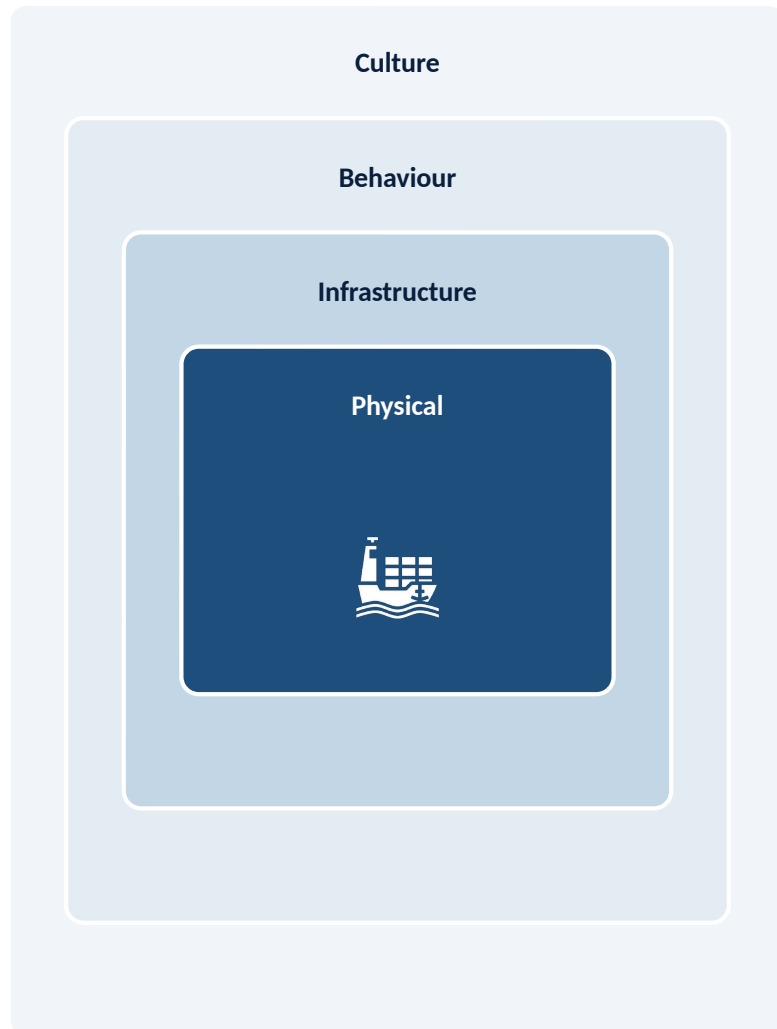
Twelve people-related factors that raise or lower the likelihood of a critical mistake being made.



Use it to mitigate risk

Knowing which factors are present in a task allows the risk to be reduced before the work starts.

A shipping company is built in four layers



Culture The foundation. Shared norms, values and perceptions that decide what is normal on board.



Behaviour What people actually do, day to day, whether or not anyone is watching.



Infrastructure The management system, procedures, resources and organisation that support the work.



Physical Ships, equipment and workplaces — the most visible layer, and the easiest to change.

Ten facts about human failure

- Human errors happen and are inevitable.
- People do not intend to do such actions, but they cannot avoid it.
- Errors are the consequences rather than the causes.
- Many errors fall into recurrent patterns.
- The same or similar events have often occurred before.
- The best people can make the worst mistakes.
- Similar types of errors occur in all industries.
- Safety failures can concern all levels of the system.
- Certain situations lead all people into the same error.
- We cannot change the human condition, but we can change the conditions in which we operate.



The conclusion that follows

If error is a consequence rather than a cause, then investigating only the person will never remove the risk. The target has to be the conditions that produced the error.



SECTION 02

The Human Factors Approach

Terms and definitions, the guiding principles, and the five pillars that hold up successful human performance.

02

Human element and human factors are not the same



Human element

The maritime term

- Used by IMO
- Recognised in many parts of the maritime industry
- Focused on changing the person to reduce human error — training, competence and motivation
- May not be recognised outside the maritime industry



Human factors (HF)

The cross-industry term

- Used across oil and gas, nuclear, aviation, space and military
- Recognises that human error is not simply a feature of individual failure
- Error is caused by workplace factors, equipment and task design, and organisational conditions
- Those conditions can lead anybody to make an error or a poor decision

The OCIMF human factors principles

- People will make mistakes.
- People's actions are rarely malicious and usually make sense to them at the time.
- Mistakes are typically due to conditions and systems that make work difficult.
- Understanding the conditions in which mistakes happen helps us prevent or correct them.
- People know the most about their work and are key to any solution.
- Plans, tools and activities can be designed to reduce mistakes and manage risk better.
- Leaders contribute in shaping conditions that influence what people do.
- It matters how leaders respond when things go wrong, and whether they take the opportunity to learn.



What this means on board

- Involve the people who do the work in designing how it is done
- Treat a mistake as information, not as a discipline case
- Look at the task and the equipment before looking at the person
- Judge leadership by its response to bad news

What the human factors approach is for



To help companies and leadership teams

Two stated aims

- Address the conditions and systems that influence human actions and decisions
- Promote safety and excellence across all operations



Organised into five focus areas

The five pillars

- Leading and shaping the culture you want
- Well-executed tasks and procedures
- Well-designed equipment and controls
- Skills to respond to emerging situations
- Learning before and after things go wrong

Five pillars of human performance

Together the five pillars support successful human performance, and through it effective, efficient, safe and secure tanker operations.



The role of leadership in defining the culture

Five leadership actions that can have a positive influence on culture.



Promote speaking up

Make it normal and safe to raise a concern.



Listen at all levels

Listening to people at every level, not only reports.



Empower to intervene

People are empowered to stop or change a task.



Communicating

Explain intent, not only instructions.



Learning vs blaming

Choose learning over attributing blame.

Well-executed tasks and procedures

Written work and real work are never identical. The gap between them is where risk lives.



What we write down

The documented system

- Procedures
- Management systems
- Risk assessments
- Work instructions
- Work permits
- Guidelines



Work as imagined vs work as done

The gap to close

Work as imagined is what the procedure assumes. Work as done is what the crew actually does, given the time, tools, weather and manning available. Where the two differ, either the procedure or the conditions must change — the gap should never simply be tolerated.

Well-designed equipment and controls

The human factors development plan should address questions such as these. If equipment can only be operated correctly by an experienced person on a good day, the design is carrying a risk that training will not remove.



Identified

Has all safety-critical equipment been identified?



Usable by everyone

Can all personnel, including those on their first work experience, operate it without error?



Reliable and maintainable

Is the equipment reliable and simple to maintain?



Placed and accessible

Is it in the proper place and easy to get at?



Suited to people

Does it make allowances for human faculties and the ways people prefer to work?



Open to alternatives

Is there new or alternative equipment that would be better?

Skills to respond to emerging situations

Crew and onshore teams often have to deal with unexpected and changing situations caused by weather, sea states, equipment and process changes and other events.



Identify and respond

Skills to identify and respond to an unexpected or changing situation as it develops, not after it has developed.



Recognise dangerous situations

The ability to see that a situation has become dangerous, even when nothing has yet gone wrong.



Maintain rarely used skills

Safety-critical skills that are seldom practised degrade quietly. They must be deliberately maintained.



Usual way or adapt?

Deciding whether to perform a task the usual way or to adapt is itself a skill, and needs judgement and support.



Do not forget mental health

Mental health is part of the capacity to respond. It belongs in the same conversation as competence and training.

Learning before and after things go wrong

The Company and its workforce should learn constantly, both from everyday work that is proceeding normally and from things that go wrong.



Key performance indicators, grouped into five

The self assessment groups its indicators into the same five areas as the five pillars.

Group	What it covers
Policy	The role of leadership in promoting a human factors approach
Capability	How the company builds its capacity to implement the policy
Design and execution	Ensuring the appropriate tools, processes and training are in place
Skills	Are staff trained to identify and respond in emerging situations?
Learning	Both when things go well and from mistakes

These five groups map directly onto the five pillars of human performance.



SECTION 03

Self-Assessment: Policy and Capability

Element 14 of the management self assessment, and the first two of its five KPI groups.

03

Four levels of maturity

Each KPI group is expressed at four levels. A company works up the levels rather than passing or failing.



Level 1

Leadership commitments. A framework to build capability. Purpose, policy and roles. The easier technical aspects.



Level 2

Beginning to implement plans, and the more complex technical aspects that would be helpful.



Level 3

Evaluating the SMS, measuring performance and improving — increasingly systematic, reliable and sustainable.



Level 4

Self-sufficiency, continuous improvement, future proofing and leading in the industry.

Policy — what the self assessment looks at

Policy covers the role of leadership in promoting a human factors approach.



Leadership essential skills

The skills leaders need in order to shape conditions rather than react to outcomes.



Human performance principles

The principles set out earlier, written into the management system and visibly applied.



Communication with the frontline workforce

Two-way communication with the people who actually carry out the work.



Organisational culture

Culture assessed through mapping methods and tools, so that it can be discussed with evidence.



Policy first

Policy is the first group because nothing else is sustainable without it. A stated commitment, backed by open dialogue between ship and shore, is what makes the remaining four groups possible.

Building the policy and opening the dialogue

Ref	Key performance indicator
14.1.1	The SMS contains a company policy on human factors along with a commitment that the company plans to implement this policy.
14.1.2	The SMS defines how the company encourages open dialogue between ship crews and shore management.
14.2.1	The human factors policy and the actions that flow from it are regularly communicated as central to the company's overall mission.
14.2.2	The company gathers information from the frontline and provides structured feedback on systems, conditions and tasks.

Policy: the role of leadership in promoting a human factors approach.

Learning through feedback, then closing the loop



Level 3

14.3.1 and 14.3.2

- 14.3.1 — Rather than looking to blame, leaders commit to learn through feedback on systems, conditions and tasks.
- 14.3.2 — The company regularly asks frontline personnel to review levels of satisfaction and trust in leadership among fleet staff.



Level 4

14.4.1 and 14.4.2

- 14.4.1 — Leaders examine how their intent and communications are received by shore-based and ship personnel, and review messaging and approach accordingly.
- 14.4.2 — The management reviews satisfaction and trust results from surveys to develop action plans and provide feedback, as needed.

Capability — what the self assessment looks at

Capability covers how the company builds its capacity to implement the policy.



Performance shaping factors

Set and measure the PSFs that shape how work is actually performed.



Define error types

A shared taxonomy so that failures can be classified consistently.



Task design

Design tasks so that the correct action is the easiest action.



Task support

Support that prevents errors and captures those that still occur.



Reducing memory load

Minimise what must be remembered; simplify the SMS accordingly.

Performance shaping factors (PSFs)

PSFs are the conditions that make correct performance more or less likely. They can be set, measured and improved.

- Fatigue, stress and boredom
- Working and living conditions
- Design and availability of equipment
- Professional knowledge and skills
- Rules, regulations and procedures
- Workload
- Distractions
- Shift patterns
- Motivation and clarity of goals
- Leadership, supervision and support
- Teamwork
- Relationship with colleagues
- National or organisational culture



Leading indicators

Because PSFs describe conditions rather than outcomes, they can be measured before an incident happens. Tracked over time they become leading indicators, giving warning while there is still time to act.

Human error taxonomy — the shape of human failure

Human failure divides into error and non-compliance. Error divides again into action and thinking.



Action

Action not as planned

- Action-based slip — the right plan, the wrong movement
- Memory-based lapse — a step forgotten or omitted
- Typically occurs in familiar, routine work



Thinking error

Action falsely planned

- Rule-based mistake — a good rule applied to the wrong situation
- Knowledge-based mistake — reasoning from incomplete understanding
- Typically occurs in novel or unclear situations



Non-compliance

Violations

- Routine — the shortcut that has become normal practice
- Situational — driven by time, tools or manning on the day
- Exceptional — a deliberate departure in unusual circumstances

From appointing owners to leading the industry

Ref	Key performance indicator
14.1.3	People are appointed to be responsible for understanding and implementing human factor policy.
14.2.3	Human factor champions are visible, conveying information and seeking input to actions related to human factors.
14.3.3	Senior shore-based managers demonstrate their understanding of human factors and engage in implementing actions covered by the human factor policy.
14.4.3	The company builds an internal human factor capability and updates its human factor policy by advocating and participating in industry forums.

Capability: how the company builds its capacity to implement the policy.



SECTION 04

Self-Assessment: Design, Skills and Learning

The three remaining KPI groups — the tools and processes, the people, and the learning that connects them.

04

Design and execution — what the self assessment looks at

Design and execution covers ensuring the appropriate tools, processes and training are in place.



Define ergonomics

A working definition of ergonomics that the fleet and shore staff can actually apply.



Record safety-critical items

Record safety-critical equipment and safety-critical tasks, so they are known and reviewed.



Impact of injuries

Understand how injuries affect the individual, the team and the operation.



Risk factor areas

Identify the areas where risk factors concentrate, and target effort there.

Assessing the conditions around safety-critical tasks

Ref	Key performance indicator
14.2.4	The company provides personnel on ships with methods, tools and training to assess conditions that shape human performance in safety-critical tasks.
14.3.4	Designated onshore managers and ship management teams regularly review the performance of safety-critical tasks.
14.4.4	Safety-critical tasks are identified across the fleet and assessed against the potential for error and performance-shaping factors.

The source table carries no Level 1 KPI for this group; assessment begins at Level 2.

Skills — what do we need to respond to emerging situations?



Resilience

The capacity of a person and a team to keep performing when the situation departs from the plan.



Soft skills against technical skills

Both are needed. Technical skill without communication and judgement is not enough in an emerging situation.



Importance of competency frameworks

A framework makes competence definable, assessable and comparable across a fleet.



Teamwork

Roles understood, workload shared, and a shared picture of what is happening.



Communication

Clear, closed-loop communication that survives noise, stress and language differences.



Rarely used, still required

Many of the skills needed in an emerging situation are used only in that situation. They have to be built and maintained deliberately, through training and scenario-based exercises.

Technical and non-technical skills

Both sets of skills are needed. Technical skill without communication and judgement is not enough when a situation departs from the plan.



Technical

Acquired and certified

- Certified knowledge
- Rules — national and international
- Industry guidelines or standards
- Customer requests and feedback



Non-technical

Personal and social

- Personal and social qualities required for certain forms of employment, which do not depend on acquired knowledge
- Identified according to the specific job profile

What human factors training covers

- Leadership and leadership styles
- Resources allocation
- Situational awareness
- Communication
- Minimisation of human error risk
- Interpersonal relations
- Management of conflicts
- Cultural differences and bias
- Team and roles
- Workload handling
- Dealing with stress, tiredness and distractions
- Drills
- How to manage people in crisis



Aimed at safety-critical roles

- Personnel in safety-critical roles are trained first
- Training is tested with scenario-based exercises
- Effectiveness is verified, not assumed
- Drills are treated as training, not paperwork

Train, exercise, then verify

Ref	Key performance indicator
14.2.5	The company provides human factors training for personnel in safety-critical roles, so they are able to respond effectively in challenging situations.
14.3.5	The company uses scenario-based exercises to assess the effectiveness of training for safety-critical activities.
14.4.5	The company verifies that training and exercises have given personnel the necessary skills to respond to emerging situations.

The source table carries no Level 1 KPI for this group; assessment begins at Level 2.

Learning — what the self assessment looks at

Learning covers both when things go well and when things go wrong.



Investigation considering HF

Incident investigation that examines human factors, not only the final action.



Proactive mentality

Looking for weak signals before an event, instead of waiting for one.



Leading indicators

Measures that give warning ahead of an incident rather than counting them afterwards.



Organisational memory

Retaining what was learned, so the same lesson is not bought twice.

From a non-blame commitment to independent investigators

Ref	Key performance indicator
14.1.4	The SMS includes commitments to cultivate a non-blame culture and to learn how human factors contribute to incidents.
14.2.6	Investigation and learning processes incorporate training and tools that enhance incident investigators' understanding of human factors.
14.3.6	Investigations are carried out with the purpose of understanding the context of actions and decisions. They avoid attributing blame, and recommend improvements to equipment, tasks, organisation and skills that are helpful to users.
14.4.6	The company builds incident investigators' competence in human factors, and demonstrates the independence of investigators' recommendations from company management.

Learning: both when things go well and from mistakes.



SECTION 05

Human Factors in the Management System

How the human factor elements sit inside the safety management system, and the cycle that keeps them improving.

05

The three human factor elements

Inside the SMS, human factors appears as culture, competency and learning.



Culture

How we treat failure

- Non-blaming
- Move from reactive to proactive, then managed, then optimised



Competency

Who does the work

- A dynamic SMS that keeps up with the operation
- Fit human factor KPIs and PSFs to the company
- Matching people to tasks
- Technical and behavioural competence together
- Understanding the types of human failure



Learning

What we do with data

- Measured KPIs used as leading indicators
- Predict and alarm before the event
- Preventive rather than corrective action

A four-phase improvement cycle

The four phases run as a continuous cycle, each one feeding the next.

1

Phase I — SMS human factors gap analysis

Optimise existing human factor components, add the missing elements, and integrate all components into a unified whole.

2

Phase II — Human factors and culture assessment

Define the performance shaping factors, identify which need improvement and which are strong, and set a PSF baseline.

3

Phase III — Performance improvement plan

Target actions to improve the PSFs, deliver customised training, and make the resulting SMS improvements.

4

Phase IV — Learn, adapt, improve

Identify trends, use PSFs as leading indicators, build corporate memory, and run a competence management system.

THE KEY MESSAGE

We cannot change the human condition, but we can change the conditions in which we operate

People will make mistakes, and their actions usually make sense to them at the time. Managing human factors means changing the conditions, the tasks and the equipment so that the right action becomes the easy action.

Change the conditions

Design out the error

Learn before it goes wrong

Four things to take away



Humans

People are the source of both error and recovery. They know the most about their work and are key to any solution.



Technology

Equipment, controls and tools either support correct performance or quietly work against it.



Performance optimisation

Measure the conditions that shape performance, and improve them deliberately rather than by reaction.



Mindset change

Move from blaming the person to understanding the context. That shift is what makes everything else possible.



Change the conditions, not just the people

Human factors belongs inside the management system: a stated policy, the capability to deliver it, well-designed tasks and equipment, the skills to respond, and constant learning.



Policy backed by open dialogue



Measure the shaping factors



Design safety-critical tasks



Learn, never blame